



KTU NOTES

The learning companion.

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NOTIFICATIONS | SOLVED QUESTION PAPERS**

Website: www.ktunotes.in

MODULE-5

Measuring classifier Performance.

Confusion Matrix

PREDICTED CLASS

TRUE CLASS	positive	negative	Total
positive	TP	FN	P
Negative	FP	TN	N

↓ (+ve)
(-ve)

p' n' N

For a TRUE CLASS problem confusion matrix is used to measure the performance of a classifier.

TP: True Positive

if the
For a positive example, prediction is also positive
Then it is called true positive.

FN: False Negative

For a positive example, if the prediction is negative
Then it is called false negative.

FP: False Positive

For a negative example, if the prediction is positive
then it is called false positive.

TN: True Negative

For a negative example, if the prediction is also
negative, it is called true negative.

Various performance measures

$$\rightarrow \text{Error} = \frac{FP + FN}{N}$$

$$\rightarrow \text{Accuracy} = 1 - \text{error}$$

$$= \frac{TP + TN}{N}$$

$$\rightarrow \text{TP rate} = \frac{TP}{P}$$

$$P = TP + FN$$

$$\rightarrow \text{TN rate} = \frac{TN}{N}$$

$$N = TP + TN$$

$$\rightarrow \text{FP rate} = \frac{FP}{n}$$

$$\rightarrow \text{Precision} = \frac{TP}{P^1}$$
$$P^1 = TP + FP$$
$$\rightarrow \text{Recall} = \frac{TP}{P}$$

$$\rightarrow \text{Sensitivity} = \frac{TP}{P}$$

$$\rightarrow \text{Specificity} = \frac{TN}{n}$$

$$n = FP + TN$$

Suppose, A computer program for recognizing dogs in a photograph identifies 8 dogs in a picture containing 12 dogs and some cats. Of the 8 dogs identified 5 actually are dogs and the rest are cats. Compute the precessional recall of the program.

True class	PREDICTED CLASS		Total
	Positive (+ve)	negative (-ve)	
+ve	TP	FN	P
-ve	FP	TN	n
	p'	n	N

Actual	Dog	Cat
Dog	TP = 5	FN = 7
Cat	FP = 3	TN =
	↓ 8	↓ 12

$$\begin{aligned}
 TP &= 5 \\
 FN &= 7 \\
 FP &= 3
 \end{aligned}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$= \frac{5}{5 + 3}$$

$$= \frac{5}{8}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$= \frac{5}{5+7}$$

$$= \frac{5}{12}$$

==

Q) Let there are 10 balls (6W + 4R) in a box and let it be required to pick up the red balls from them.

Suppose we pick up 7 balls as the Red balls of which only two are actually red balls. What are the precision and recall in picking a Red ball.

	PR	FW	
AR	2	2	4
AW	5	1	6
	7	3	10

2		
		10

$$7-2=5$$

$$4-2=2$$

$$6-5=1$$

$$7+3=10$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$= \frac{2}{2+5}$$

$$= \frac{2}{7}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\frac{\frac{2}{2+7}}{\frac{2}{2+7}} = \frac{2}{2+2}$$

e) Calculate F-score = $\frac{2}{\frac{1}{\text{Precision}} + \frac{1}{\text{Recall}}}$

$$= \frac{2}{\frac{1}{\frac{2}{7}} + \frac{1}{\frac{1}{2}}} = \frac{4}{11}$$

Q) Assume the following.

- * A database contains 80 records on a particular topic of which 55 are relevant to a certain investigation.
- * A search was conducted on that topic and 50 records were retrieved.
- * Of the 50 records retrieved 40 were relevant construct the confusion matrix for the search and calculate precision and recall.

	PR	PNR	
Actual Relevant	40 TP	15 FN	55
Actual Not Relevant	10 FP	15 TN	25
	50	30	80

$$\begin{aligned}\text{Precision} &= \frac{TP}{TP+FP} \\ &= \frac{40}{40+10} \\ &= \frac{40}{50}\end{aligned}$$

$$\begin{aligned}\text{Recall} &= \frac{TP}{TP+FN} \\ &= \frac{40}{40+15} \\ &= \frac{40}{55}\end{aligned}$$

Q) Suppose 10,000 patients get tested for Covid out of them 9000 are actually healthy [not positive] and 1000 are actually sick. [positive]

For the sick people a test was positive for 620 and negative was 380

For healthy people the same test was positive for 180 and negative for 8820.

Construct a confusion matrix for the above data and calculate the precision recall.

	PH	PNH	
Actually healthy	8820	180	9000
Actually not healthy	380	620	1000
			10000

ROC

* Receiver Operating Characteristics.

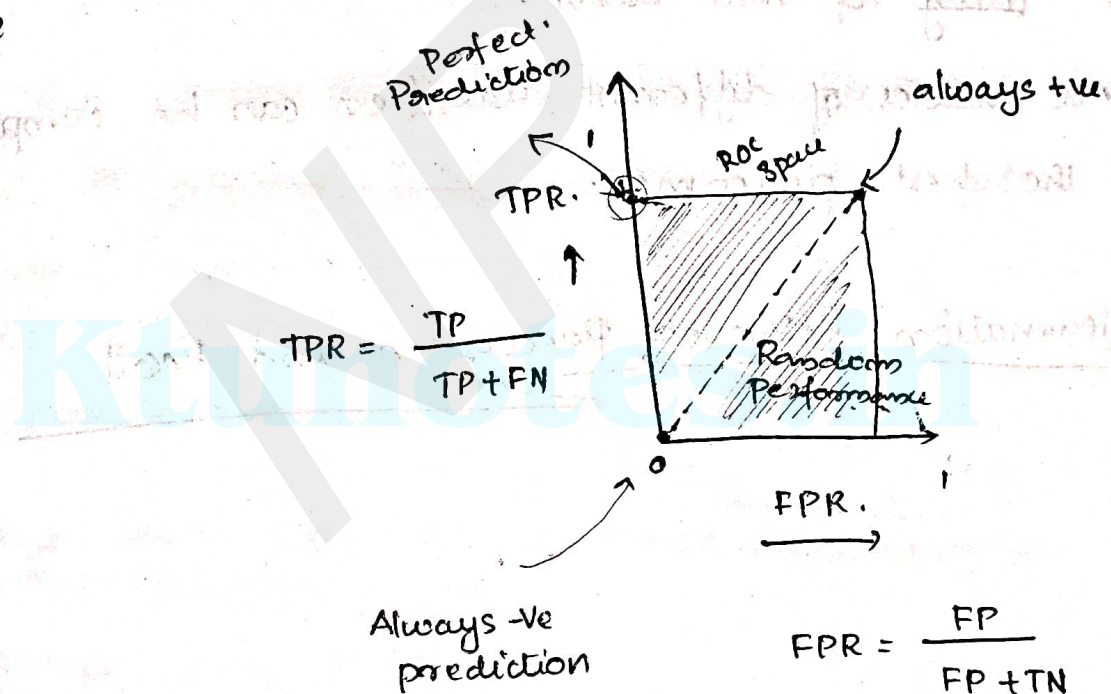
* It is used for visualising the performance.

* It is used in machine learning and data mining

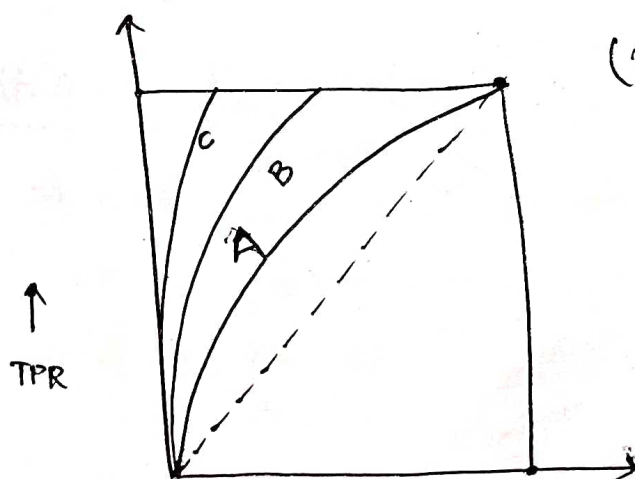
* ROC curve is obtained by plotting the values of

FPR ~~ROC space~~ [False Positive Rate]

and the value of ~~FP~~ TPR along the Y axis in a plane



example :-



A → classifiers

(ROC curves of classifiers A, B, C)

AUC → Area Under the Curve.

Here,

classifier C is closest to the top left corner of the Roc space.

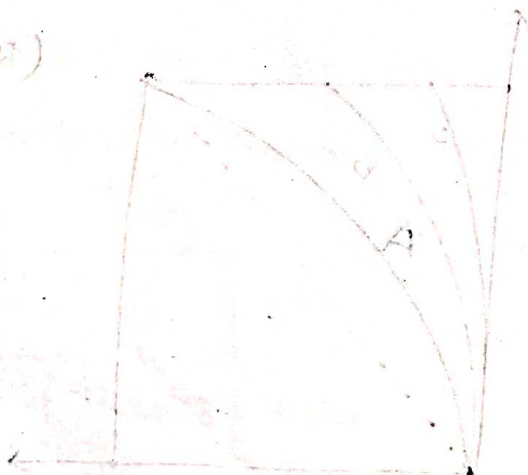
Among the 3 classifiers C is the best classifier.

AUC - Area Under the Curve

If you want to reduce the curve to a single number we can do it by calculating area under the curve using of that curve.

AUC values of different classifiers can be compared to get find the best performer.

Information Retriever Precision and Recall



Combining Multiple learners (Ensembling)

- * None of the single learning algorithm is ~~not~~ perfect
- * It is possible to create models that composed of multiple learners that complement each other.
- * Model combination does always increase time and space complexity of training and testing
- * So, base learners should be trained carefully and the decisions should be combined efficiently else accuracy will reduce and complexity will increase.
- * There are two issues in combining multiple learners
 1. How do we generate base learners that complement each other
 2. How do we combine the outputs of base learners for maximum accuracy.

Generating base learners ^(Generating diverse base learners)

1* Use different algorithms

We can use different learning algorithms to train different base learners [example SVM, decision tree, Neural network etc]

2* Use different hyper parameters

eg:- SVM $\rightarrow$ Parameter kernel function.

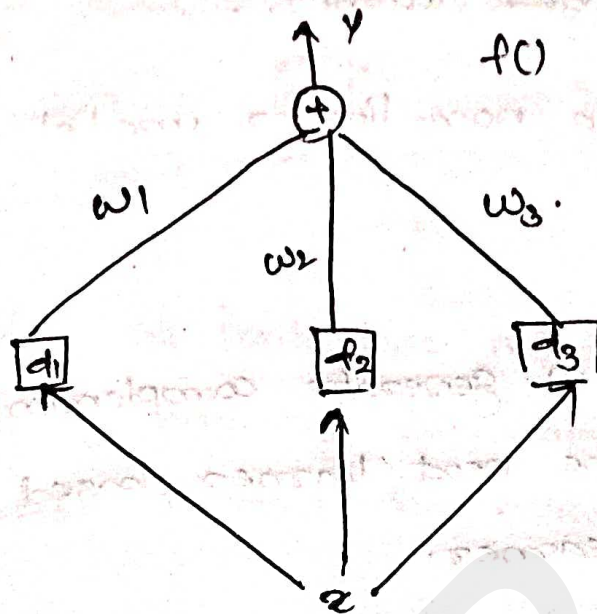
DT $\rightarrow$ No. of nodes, depth

NN $\rightarrow$ No. of neurons, no. of hidden layers ~~etc~~
No. of weighted connections to bias neuron

3* Use different input representation

For example in speech recognition to recognize

VOTING

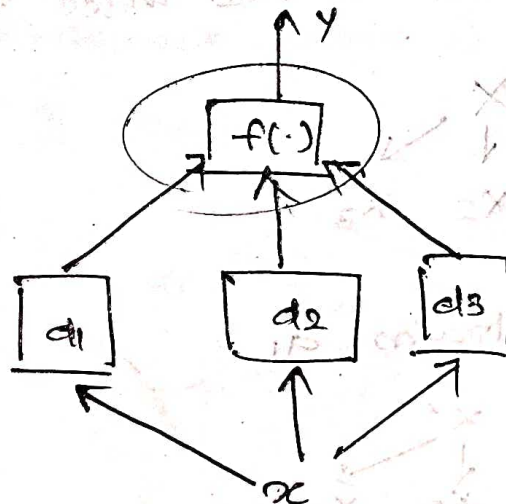


$$y = \sum_j w_j d_j$$

when $w_j \geq 0$

$\sum_j w_j = 1$ & $d_j \Rightarrow$ o/p produced by model of 'j'.

STACKING



- * Combiner is another learner
- * It can learn how the base learners make errors so it can correct the ^{o/p of} base learners.
- * Combine $f(i)$ can be a non-linear model

BOOSTING

In boosting we try to generate complementary base learners. By training the next learner based on the mistakes of previous learner.

Original boosting algorithm uses three weak learners to generate a strong learner.

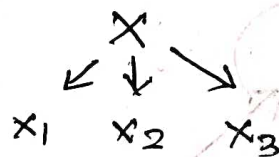
Weak learner:- It has error probability less than $1/2$.

Strong learner:- It has very small error probability.

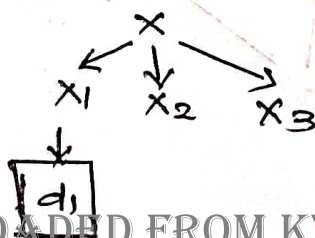
Method

Given a training set X

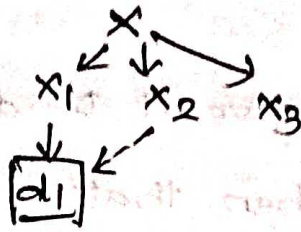
We randomly divide it into three subsets.



We use X_1 and train d_1

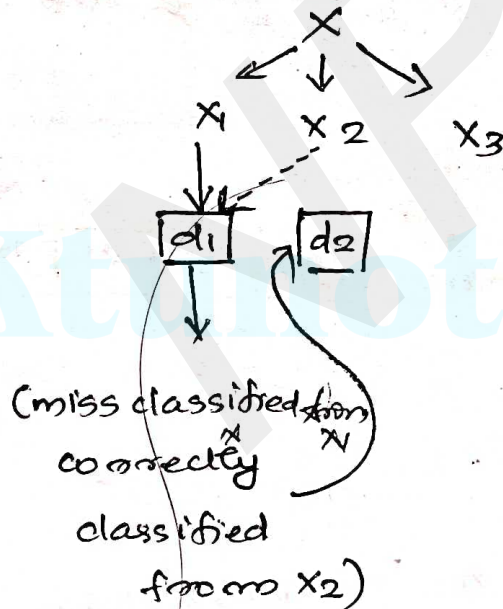


then take X_2 then feed it to d_1 for testing

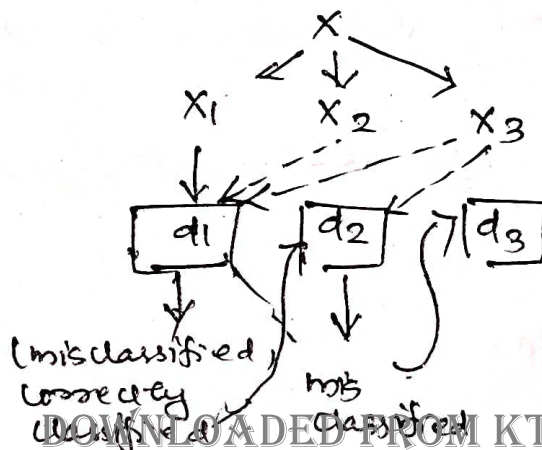


Then,

We take all instances misclassified by d_1 and also instances those correctly classified from X_2 . These together formed the training set of d_2 .



We then take X_3 and feed it to d_1 and d_2 misclassified instances from d_1 and d_2 form the training set of d_3

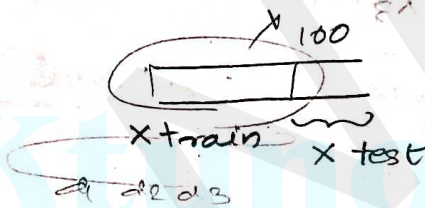


Given an instance we give it to d_1 and d_2 . If they agree [same Result] Then that is the final response else, Response of d_3 is taken as final output.

Monday
28/11/2022

Cross validation

Assume that we have enough training and validation set available



Finding a hypothesis -
hy.

That is most accurate on the validation set is called cross validation

imp Methods used for cross validation

1) K-fold cross validation

In a K-fold cross validation is iterate.

Is divided randomly into K-equal sized part

K-fold (5-fold)

$$X = 100$$

20	20	20	20	20
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then, we keep one of the k -part as the validation set and combine remaining $k-1$ parts to form the training set.

Then the model is created using $k-1$ parts error of the model is measured using one of the k -part. the above process is repeated k -times

eg:-

Suppose $X = 100$ & $k = 5$

Iteration 1

X_1	X_2	X_3	X_4	X_5
20	20	20	20	20

$$V_1 = X_1 \quad T_1 = X_2 \cup X_3 \cup X_4 \cup X_5$$

$$E(V_1) =$$

Iteration 2

X_1	X_2	X_3	X_4	X_5
20	20	20	20	20

$$V_2 = X_2$$

$$T_2 = X_1 \cup X_3 \cup X_4 \cup X_5$$

$$E(V_2) =$$

5 times repeat above process
accuracy value

Iteration 3.

x_1	x_2	x_3	x_4	x_5
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$$V_1 = x_3$$

$$T_3 = x_1 \cup x_2 \cup x_4 \cup x_5$$

$$E(V_3) =$$

Iteration 4

x_1	x_2	x_3	x_4	x_5
-------	-------	-------	-------	-------

$$V_1 = x_3$$

$$T_4 = x_1 \cup x_2 \cup x_3 \cup x_5$$

$$E(V_4) =$$

Iteration 5

x_1	x_2	x_3	x_4	x_5
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$$V_1 = x_4$$

$$T_5 = x_1 \cup x_2 \cup x_3 \cup x_4$$

$$E(V_5) =$$

Final accuracy of the model

$$E(V) = \frac{E(V_1) + E(V_2) + E(V_3) + E(V_4) + E(V_5)}{5}$$

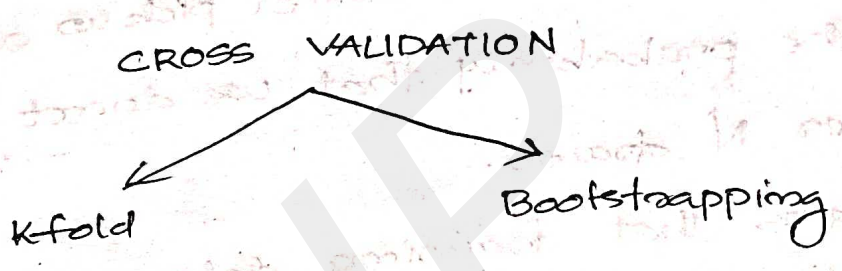
Tuesday
29/11/2022

PROBLEMS (DISADVANTAGES)

- * Training set overlap [Any two training set share $K-2$ part]
- * To keep the training set large validation set should be ^{set} small

what will happen when k becomes large.
 As k increases the percentage of training instances increases. and we get good estimators or models.
 But the validation set become smaller.

Leave One Out 1. One extreme case of k -fold cross validation is leave one-out where given a dataset of n instances, only one instance is left out as the validation instance. And the training uses $n-1$ instances.



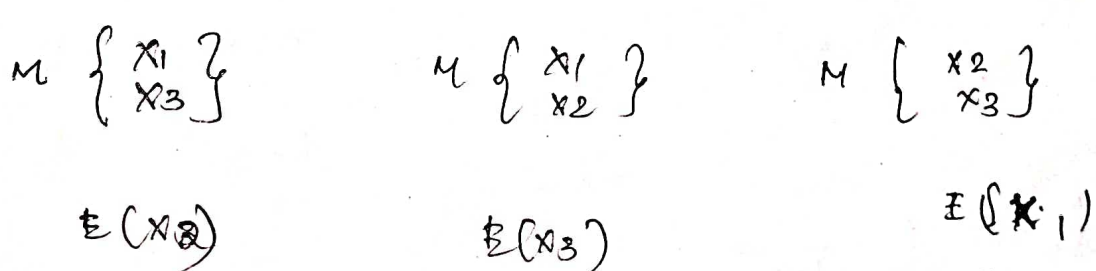
Bootstrapping

It is an alternative to k -fold cross validation

- * It is used to generate multiple samples from a given sample
- * It uses the concept of probability.
- * It generate new samples by the method of "drawing i"
 - "taking instances from the original's with replacement."

eg:- $\{x_1, x_2, x_3\}$

$N=3$
 $M=2$



In the bootstrap method we sample m instances from a datasets of size N with replacement. Then the ~~original dataset~~ model is created using ~~the~~ m instances then original dataset used as validation set.

* The above process is repeated for N times

disadvantage

The probability that we pickup an instance from N instances is $1/N$ so we do not pickup an instance is $1 - \frac{1}{N}$. Then the probability that we do not pickup an instance after N draws.

This means that training data contains approximately 63.2% instances. That is the system will not have been trained on 36.8% data

Replication

The Above problem can be solved by using replication. That is repeat the process of bootstrapping many times and take the average error.

e